

User Input (Scanner)

Lernziele

Verschiedene Methoden zum Einlesen

Beispiel Scanner

Methoden für spezifische Datentypen

Aufgaben

Lernziele

- Benutzereingabe mithilfe der Scanner Klasse lesen und ausgeben können

Benutzereingabe einlesen

- Verschiedene Methoden
 - Buffered Reader Class
 - Console Class
 - **Scanner Class**
 - Command Line Arguments → Nicht für das Gedacht!

Scanner Class Beispiel

```
public class Main {  
    public static void main(String[] args) {
```

```
    }  
}
```

Scanner Class Beispiel

```
public class Main {  
    public static void main(String[] args) {  
  
        System.out.print("Enter your name: ");  
  
        // Print out the name  
        System.out.println("Your name is " + name);  
  
    }  
}
```

Scanner Class Beispiel

```
public class Main {  
    public static void main(String[] args) {  
  
        System.out.print("Enter your name: ");  
  
        String name  
  
        // Print out the name  
        System.out.println("Your name is " + name);  
  
    }  
}
```

Scanner Class Beispiel

```
public class Main {  
    public static void main(String[] args) {  
  
        // Create an object of the scanner class  
        Scanner userInput = new Scanner(System.in);  
  
        System.out.print("Enter your name: ");  
  
        String name  
  
        // Print out the name  
        System.out.println("Your name is " + name);  
  
    }  
}
```

Scanner Class Beispiel

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        // Create an object of the scanner class
        Scanner userInput = new Scanner(System.in);

        System.out.print("Enter your name: ");

        String name

        // Print out the name
        System.out.println("Your name is " + name);

    }
}
```

Scanner Class Beispiel

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        // Create an object of the scanner class
        Scanner userInput = new Scanner(System.in);

        System.out.print("Enter your name: ");

        // Read the input from the Keyboard and store it in the variable "name"
        String name = userInput.nextLine();

        // Print out the name
        System.out.println("Your name is " + name);

    }
}
```

Scanner Class Beispiel

```
import java.util.Scanner; ← Dont forget the import

public class Main {
    public static void main(String[] args) {

        // Create an object of the scanner class
        Scanner userInput = new Scanner(System.in);

        System.out.print("Enter your name: ");

        // Read the input from the Keyboard and store it in the variable "name"
        String name = userInput.nextLine();

        // Print out the name
        System.out.println("Your name is " + name);

        // Close the scanner
        userInput.close(); ← Garbage Collector
    }
}
```

Datentypen lesen

- Methoden zum Einlesen der Datentypen

Datentyp	Beschreibung	Method
Byte,	Ganzzahl	nextByte()
Short,	Ganzzahl	nextShort()
Int,	Ganzzahl	nextInt()
Long,	Ganzzahl	nextLong()
Float,	Gleitkommazahl	nextFloat()
Double,	Gleitkommazahl	nextDouble()
Boolean,	Warheitswert	nextBoolean()
String,	Text	nextLine()
Char,	einzelnes Zeichen	next().charAt(n); // n = an welcher Stelle, index an n-ter Stelle

Achtung

```
Scanner userInput = new Scanner(System.in);
```

```
System.out.println("The sum of 1+1 is: ");
```

```
byte result = userInput.nextByte(); // 2
```

```
System.out.println("Your answer is: " + result);
```

```
System.out.println();
```

```
System.out.println("What is your name? ");
```

```
String userName = userInput.nextLine(); // Jony
```

```
System.out.println("Your name is " + userName);
```

```
userInput.close();
```



```
The sum of 1+1 is:  
2  
Your answer is: 2  
  
What is your name?  
Your name is  
  
Process finished with exit code 0
```

Achtung

```
Scanner userInput = new Scanner(System.in);

System.out.println("The sum of 1+1 is: ");
byte result = userInput.nextByte(); // 2
System.out.println("Your answer is: " + result);

System.out.println();

System.out.println("What is your name?");
userInput.nextLine();
String userName = userInput.nextLine(); // Jony
System.out.println("Your name is " + userName);

userInput.close();
```

Advances this scanner past the current line and returns the input that was skipped.

<https://docs.oracle.com/en/java/javase/22/docs/api/java.base/java/util/Scanner.html>

Nach der Verwendung von `nextInt()` oder anderen `next`-Methoden (ausser `nextLine()`), sollten wir ein zusätzliches `nextLine()` einfügen, um den verbleibenden Zeilenumbruch(`\n`) zu konsumieren

```
The sum of 1+1 is: 2
Your answer is: 2

What is your name?Jony
Your name is Jony

Process finished with exit code 0
```



Hinweis

- Java 25+ bietet neu die IO Klasse an, welche **print**, **println** und **readln** Methoden anbietet.
- Dies macht das Einlesen von Benutzereingaben einfacher:

```
// OLD  
Scanner userInput = new Scanner(System.in);  
System.out.print("Enter your name: ");  
String inputOld = userInput.nextLine();  
System.out.println("Your name is " + inputOld);
```

```
// NEW  
String inputNew = IO.readln("Enter your name: ");  
IO.println("Your name is " + inputNew);
```

Fragen?

Aufgaben

- InputEinlesen